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|--|--|----------------------|-------------------|
| 4 | <p>(a) wave incident on/passes by or through an aperture/edge
wave spreads (into geometrical shadow)</p> | B1
B1 | [2] |
| <p>(b) (i) waves (from slits) overlap (at point X)</p> <p style="padding-left: 40px;">path difference (from slits to X) is zero/
phase difference (between the two waves) is zero
(so constructive interference gives bright fringe)</p> | | B1 | |
| <p>(ii) difference in distances = $\lambda/2 = 580/2$
= 290 nm</p> | | B1 | [2] |
| <p>(iii) $\lambda = ax/D$</p> <p style="padding-left: 40px;">$D = [0.41 \times 10^{-3} \times (2 \times 2.0 \times 10^{-3})]/580 \times 10^{-9}$</p> <p style="padding-left: 40px;">= 2.8 m</p> | | A1
C1
C1
A1 | [1]
[2]
[3] |
| <p>(iv) same separation/fringe width/number of fringes
bright fringe(s)/central bright fringe/(fringe at) X less bright
dark fringe(s)/(fringe at) Y/(fringe at) Z brighter
contrast between fringes decreases</p> <p style="padding-left: 40px;"><i>Any two of the above four points, 1 mark each</i></p> | | B2 | [2] |

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